

Help Desk Automation Orchestrator

Runbook and Switch Reference

Item	Value
Package	Invoke-HelpDeskOrchestrator.ps1, Invoke-HelpDeskRollback.ps1
State Path	C:\ProgramData\HelpDeskAutomation\State\Operations
Log Path	C:\ProgramData\HelpDeskAutomation\Logs
Version	2.0

1. Prerequisites

Requirement	Detail
PowerShell	5.1 or later
Execution context	Run as Administrator. Does not self-elevate to SYSTEM.
Execution policy	Must allow script execution. Use <code>Set-ExecutionPolicy Bypass -Scope Process</code> if needed.
OS	Windows 10 / Windows 11 Desktop. Not validated on Server.
File placement	Keep both scripts in the same folder.
Signed scripts	If your environment enforces signing or Constrained Language Mode, sign the scripts before deployment.

2. Operating Model

- Nothing runs unless a remediation switch is explicitly selected.
- Irreversible actions are blocked unless `-AllowIrreversible` is supplied.
- Every run writes timestamped logs and an operation snapshot for rollback.
- Rollback is available only for modules that capture restorable state.
- Profile health is audit-only by design.

3. Quick Start

```
.\Invoke-HelpDeskOrchestrator.ps1 -RepairWindowsUpdate -ClearLegacyWsusPolicy -ResetWindowsUpdateWin  
.\Invoke-HelpDeskRollback.ps1 -UseLatestOperation  
.\Invoke-HelpDeskRollback.ps1 -OperationId OP-20260331_153000-AB12CD34
```

4. Files and Default Locations

Item	Purpose	Default Location
Orchestrator	Primary entry point for approved remediations	Package folder
Rollback script	Emergency backout for reversible changes	Package folder
Logs	Per-run log files	C:\ProgramData\HelpDeskAutomation\Logs
Operation snapshots	Rollback metadata and backup state	C:\ProgramData\HelpDeskAutomation\State\Operations
Baseline runner	Script created by the weekly scheduled task	C:\ProgramData\HelpDeskAutomation\Scripts
Print quarantine	Spool files moved out of the live queue	C:\ProgramData\HelpDeskAutomation\PrintSpoolerQuarantine

5. Log Format

Each run creates a log file named `{OperationId}.log`. Format:

```
[2026-04-01 14:30:00] [INFO] === MODULE: RepairWindowsUpdate ===
[2026-04-01 14:30:01] [INFO] Stopping service: wuauserv
[2026-04-01 14:30:02] [INFO] Backing up SoftwareDistribution
[2026-04-01 14:30:05] [SUCCESS] RepairWindowsUpdate completed successfully
```

Levels: **INFO** (normal step), **SUCCESS** (module completed), **WARN** (action completed but needs attention), **ERROR** (module failed), **AUDIT** (profile health finding, no changes made).

The OperationId is printed at the start and end of every run. It is also the filename of both the log and the snapshot JSON.

6. Control and Rollback Switches

Switch	Used with	What it does
-AllowIrreversible	Orchestrator	Unlocks guarded actions without reliable rollback
-BaselineTaskName	Orchestrator	Overrides the scheduled task name
-BaselineDayOfWeek	Orchestrator	Sets the weekly day for the baseline task
-BaselineTime	Orchestrator	Sets the task start time
-OperationId	Rollback	Targets a specific operation snapshot

Switch	Used with	What it does
-UseLatestOperation	Rollback	Targets the most recent operation snapshot
-Modules	Rollback	Limits rollback to specific module names

Accepted rollback module names: WindowsUpdate, PrintSpooler, TimeService, NetworkIdentity, Defender, DiskPressure, WeeklyBaselineRegister.

7. Primary Remediation Switches

Switch	What it targets	Common modifiers	Rollback coverage
-RepairWindowsUpdate	Update scan hangs, WSUS residue, cache corruption	-ClearLegacyWsusPolicy, -ResetWindowsUpdateWinHtpProxy	Restores WSUS registry values, WinHTTP proxy, and cache backups
-RepairPrintSpooler	Stuck print jobs, spool queue corruption	None required	Restores spooler startup mode and quarantined spool files
-RepairTimeService	Clock skew, domain hierarchy resync	-ManualPeerList	Restores W32Time startup mode and prior config
-RepairNetworkIdentity	DNS, DHCP, WinHTTP proxy issues	-RenewDhcpLease, -ResetNetworkWinHttpProxy	Restores WinHTTP proxy only
-RepairDefender	Signature drift, update source validation	-DefenderSignatureFallbackOrder, -DefenderUpdateSource, -DefenderScheduleDailyCheck	Restores Defender update preferences
-AuditProfileHealth	TEMP profiles, orphaned SIDs, .bak entries	-DeleteProfilesUnusedForDays	N/A (audit-only, no changes made)
-RelieveDiskPressure	Low free space, temp bloat, hibernation	-CleanupTemp, -RunComponentCleanup, -DisableHibernation	Re-enables hibernation only
-RegisterWeeklyBaseline	Scheduled maintenance task	-BaselineTaskName, -BaselineDayOfWeek, -BaselineTime	Unregisters the scheduled task

8. Irreversible Actions

These require **-AllowIrreversible**:

-RunComponentCleanup — Runs **DISM /StartComponentCleanup /ResetBase**. Compresses the WinSxS component store and permanently removes superseded updates. After this runs, those updates cannot be uninstalled. Use only when disk space is critically low and the endpoint is stable on its current update level.

9. Standard Workflows

Windows Update failure

```
.\Invoke-HelpDeskOrchestrator.ps1 -RepairWindowsUpdate -ClearLegacyWsusPolicy -ResetWindowsUpdateWin
```

Print queue stuck

```
.\Invoke-HelpDeskOrchestrator.ps1 -RepairPrintSpooler
```

Clock skew / authentication issue

```
.\Invoke-HelpDeskOrchestrator.ps1 -RepairTimeService
```

DNS / proxy / stale registration

```
.\Invoke-HelpDeskOrchestrator.ps1 -RepairNetworkIdentity -RenewDhcpLease -ResetNetworkWinHttpProxy
```

Defender health drift

```
.\Invoke-HelpDeskOrchestrator.ps1 -RepairDefender
```

Low disk space

```
.\Invoke-HelpDeskOrchestrator.ps1 -RelieveDiskPressure -CleanupTemp -DisableHibernation
```

Weekly baseline deployment

```
.\Invoke-HelpDeskOrchestrator.ps1 -RegisterWeeklyBaseline -BaselineDayOfWeek Sunday -BaselineTime 03
```

Profile health check

```
.\Invoke-HelpDeskOrchestrator.ps1 -AuditProfileHealth -DeleteProfilesUnusedForDays 60
```

Audit-only. Review output, then follow manual remediation steps in Section 12.

10. Rollback Procedure

1. Find the OperationId in the orchestrator output or log filename.
2. Run the rollback script:

```
# Roll back everything from a specific operation
.\Invoke-HelpDeskRollback.ps1 -OperationId OP-20260401_143000-A1B2C3D4

# Roll back only specific modules
.\Invoke-HelpDeskRollback.ps1 -OperationId OP-20260401_143000-A1B2C3D4 -Modules WindowsUpdate,PrintS

# Roll back the most recent operation
.\Invoke-HelpDeskRollback.ps1 -UseLatestOperation
```

- 3. Validate service state and functionality after rollback.

Partial Failure Handling

If the orchestrator runs multiple modules and one fails, the snapshot still captures rollback data for all modules that ran. You can roll back modules independently using `-Modules`. A failure in one module does not corrupt rollback data for other modules.

11. Rollback Matrix

Module	Can restore	Cannot restore
WindowsUpdate	SoftwareDistribution cache, catroot2 cache, WSUS registry values, WinHTTP proxy	Re-registered DLLs (harmless)
PrintSpooler	Quarantined spool files, startup mode	N/A
TimeService	Startup mode, partial config	Manual peer list may need reconfiguration
NetworkIdentity	WinHTTP proxy	DNS flush and DHCP renew are non-destructive
Defender	Signature fallback order, update preferences	Forced signature update stays (not harmful)
ProfileHealth	N/A (audit-only)	N/A
DiskPressure	Hibernation on/off	Deleted temp files, DISM component cleanup
WeeklyBaseline	Unregisters scheduled task	N/A

12. Post-Audit Remediation: Profile Health

When `-AuditProfileHealth` reports findings, these manual steps are recommended. These are intentionally not automated because they involve destructive registry changes.

.bak Registry Entries (Duplicate SID)

- 1. Open Registry Editor → `HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\ProfileList`.
- 2. Find the SID with `.bak` suffix and the duplicate without `.bak`.
- 3. Rename the non-.bak entry to `.old`, then remove `.bak` from the original entry.

4. Reboot and verify.

TEMP Profile Accumulation

1. Log the affected user off completely.
2. Delete `C:\Users\TEMP.*` folders.
3. Fix the underlying .bak entry (see above).
4. Reboot and verify.

Orphaned Registry Entries

1. Back up the registry key.
2. Delete the orphaned SID key from ProfileList.

Stale Profiles

1. Confirm with user/manager that the profile is no longer needed.
2. Remove with `Get-CimInstance Win32_UserProfile | Where-Object { $_.LocalPath -eq 'C:\Users\USERNAME' } | Remove-CimInstance`.

Escalation: If more than 5 profiles are affected or 10+ TEMP folders exist, escalate before manual remediation. This pattern often indicates a deeper domain trust, GPO, or SID replication issue.

13. Escalation Rule

Do not stack heavy remediations on the same endpoint. If one controlled module does not solve the issue, stop and escalate rather than turning the machine into a moving target.